SignPlug - ISL plugin for the web

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Abstract

We present SignPlug, a lightweight, real time framework for synthesizing continuous Indian Sign Language (ISL) avatar motion from input sentences directly on the web. The system is designed to bridge the communication gap for Deaf and hard-of-hearing users. Digital accessibility research has grown substantially, yet native browser support for sign language rendering remains largely absent. Existing signing avatar pipelines are either motion capture systems too costly to scale, GPU hungry neural motion generators, or notation driven systems such as HamNoSys and SiGML that demand substantial linguistic expertise and yield mechanical, unnatural movement. None of these reconstruct motion directly from real signers at dictionary scale.

We address this gap by reconstructing SMPL-X keyposes directly from monocular ISLRTC ISL dictionary videos and refining them with a purpose-built sparse correction annotation tool, Gizmo. This yields the Widget Dictionary of reusable, manually verified keyposes with corrected body, hand, and finger articulation. At inference time, glosses predicted from an input sentence are retrieved from this dictionary, retargeted onto a canonical SMPL-X avatar, and composed via spherical linear interpolation (SLERP), avoiding both frame by frame animation and GPU based motion generation entirely. SignPlug therefore enables low latency, browser deployable ISL sentence synthesis grounded in captured human motion rather than symbolic notation. We evaluate SignPlug through synthesis latency, playback frame rate, gloss retrieval against real signer videos, and a study with 11 Deaf and hard-of-hearing ISL signers, who identified 88.8% of avatar sentences correctly. These results position SignPlug as a step toward scalable, natural 3D sign language accessibility for Deaf and hard-of-hearing communities and everyone who interacts with them.

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CCS Concepts

• **Human-centered computing** → **Accessibility systems and tools**; • **Computing methodologies** → *Animation*; *Computer vision*; Natural language processing.

Keywords

Indian Sign Language, sign language avatar, SMPL-X, gloss dictionary, pose correction, real-time animation, accessibility

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Figure 1: Rendered playback of a full ISL sentence, "He wants to eat now," on the SignPlug avatar, generated end-to-end by the pipeline described in Section 3. View in Adobe Reader to play the animation.

1 Introduction

> *"The power of the Web is in its universality. Access by everyone regardless of disability is an essential aspect."*
>
> –Tim Berners-Lee, inventor of the World Wide Web

Hearing loss remains one of the most under addressed disabilities worldwide. The World Health Organization estimates that over 1.5 billion people currently live with some degree of hearing loss, of whom roughly 430 million experience disabling hearing loss that requires rehabilitation, a number projected to exceed 700 million by



Figure 2: Demo of the SignPlug widget embedded in a locally hosted ICVGIP website, showing the avatar rendering the sentence "He wants to go to the office" from real time gloss retrieval and playback.

2050 [9]. A large fraction of this population communicates primarily through one of the more than 300 sign languages (SLs) in use around the world [54]. India alone is home to an estimated 18 to 63 million Deaf and hard-of-hearing individuals [22], who rely on Indian Sign Language (ISL), linguistically distinct from spoken Hindi or English, as their primary means of communication. Despite this scale, accurate generative models of holistic 3D signing avatars, together with large, well annotated dictionaries, remain highly desirable but scarcely available for efficient SL learning, teaching, and translation [37].

Sign languages are complete, independent natural languages with their own syntax and grammar. They rely on spatial visual articulation rather than acoustic signals. Sign languages are not universal. Distinct Deaf communities use distinct sign languages. ISL is distinct from ASL and BSL in vocabulary and grammar, so models trained on other sign languages do not transfer directly. Prior computational ISL systems often model gloss order in subject-object-verb (SOV) form, whereas ASL is widely described as basic subject-verb-object (SVO). Sign language word order, however, is flexible and construction dependent [40, 44]. Region specific 3D motion capture and avatar generation are therefore essential for local accessibility.

Modern web platforms routinely accommodate users with color blindness, low vision, or motor impairments, but comparable native support for Deaf users, in the form of sign language rendering rather than text captions alone, remains largely absent. Closing this gap requires more than 2D video playback, since 2D motion sequences are known to be insufficient for reliable SL learning and recognition, as distinct signs can project onto visually indistinguishable 2D silhouettes due to depth ambiguity [52]. Faithful, learnable, and reproducible signing therefore requires 3D holistic avatar motion, with accurate body, hand, and facial articulation reconstructed or generated in three dimensions. Beyond avatar generation, 3D pose annotations also drive sign language recognition and translation systems that operate directly on video [18, 30, 41, 53], accessibility and education tools such as interactive sign dictionaries and real time speech to avatar systems [4, 7, 13], immersive AR and VR signing environments [57], and monocular 3D reconstruction pipelines that grow new datasets without costly motion capture rigs [1, 5, 32].

Three compounding obstacles stand in the way of 3D ISL avatar generation. First, while parametric holistic body models are well established for generic human motion [42, 55], no existing dataset provides ISL motion together with accurate 3D hand mesh annotations, a level of detail indispensable for accurate handshapes. Second, fully hand crafted avatar pipelines, in which animators or SL experts manually author each sign, are labor intensive and fundamentally unscalable to dictionary sized vocabularies. Third, notation driven systems that generate motion from HamNoSys or similar symbolic annotations require substantial linguistic expertise to author and typically yield robotic, unnatural movement, since the resulting motion is synthesized from a symbolic intermediate representation rather than captured directly from human signers.

SignPlug builds on established tools such as SMPLer-X and SLERP. Its novelty lies in what they produce together. We make two contributions:

- **Widget Dictionary** (Section 3.1): manually corrected SMPL-X keyposes for ISL glosses, built from ISLRTC dictionary videos. To our knowledge, no manually verified 3D equivalent existed for ISL before. Existing sign language production datasets lack such keyframe annotations [34].
- **SignPlug web plugin** (Section 3): it synthesizes ISL sentences entirely in the browser, with no GPU or server-side model, at 25 FPS and 0.13–1.42 s preparation time for 2–20 glosses.

2 Related Work

A separate line of work targets the complementary sub-problem of converting an English sentence into its corresponding ISL gloss sequence. We do not propose a new text-to-gloss model; instead, we obtain gloss sequences from an LLM (Section 3) and focus on generating motion from them. Nair and Idicula [38] address gloss generation as a low-resource machine-translation task. Their method uses a constituency parse tree grammatical transformation followed by rule-based mapping to ISL gloss structure. The same group later replaced the rule-based reordering stage with a BiLSTM encoder and LSTM decoder sequence-to-sequence model. This change reported improved BLEU and ROUGE-L scores over the rule-based system [39]. At the dataset level, ISLTranslate [25] and iSign [27] provide large parallel English-ISL sentence, gloss, and pose corpora. Such gloss-generation systems are trained and evaluated on these corpora. CISLR [26] supplies the word and gloss-level recognition resource many of these pipelines rely on for gloss vocabulary grounding. Sugandhi et al. [48] similarly take English or Hindi text as input, but target a HamNoSys intermediate representation rather than a plain gloss sequence. Their system drives a hand-crafted avatar directly from the parsed notation. In contrast, we focus on the downstream problem of producing natural, co-articulated 3D motion from the resulting gloss sequence.

Recovering holistic 3D human-body avatars from RGB video and parsing them into parametric forms such as SMPL-X [42] is a well-explored problem for general human motion capture. Recent expressive regressors such as SMPLer-X [6] have scaled this estimation to in-the-wild video. However, such parametric holistic models have rarely been applied to the sign language domain. The closest prior work is Forte et al. [13], who reconstruct 3D holistic signing avatars from video using linguistic priors. They obtain group labels from a sign classifier trained on the Corpus-based Dictionary of Polish Sign Language (CDPSL) [33], itself annotated with HamNoSys. Their method relies on an existing sentence-segmentation approach [43] to generalise beyond isolated signs. Their pipeline, driven by a HamNoSys classifier, is cost-efficient but does not scale gracefully with vocabulary size. To the best of our knowledge, no equivalent 3D holistic reconstruction pipeline has been proposed for Indian Sign Language specifically.

Public sign language datasets remain dominated by RGB video and 2D keypoints. This is despite the fact that 2D representations lose depth and can collapse distinct 3D hand configurations into similar projections [18, 32, 57]. This matters for sign production because realistic signing depends on coordinated body, hand, face, and non-manual articulators rather than hand motion alone [35, 49]. For ISL specifically, available datasets largely support recognition or translation from video or landmarks rather than holistic 3D avatar generation. Recent ISL work introduces continuous and isolated corpora for recognition, but these remain video-based or pose-based rather than parametric 3D body models [14, 23, 36]. Several ISL systems therefore rely on extracted landmarks or keypoints, since collecting richer training data remains difficult [8, 28, 36]. Outside ISL, recent 3D sign language resources have started to address this gap. SignAvatars [57] introduces a large-scale 3D holistic dataset with continuous and co-articulated signs, precise hand meshes, and multimodal prompts. Neural Sign Actors [2] extends How2Sign with SMPL-X annotations for large-scale 3D production research, but neither targets ISL. Recent efforts in other sign languages, including Arabic Sign Language, make a similar point. 3D parametric annotations form foundational infrastructure for reconstruction, generation, and evaluation, and their absence limits avatar-based downstream systems [1]. Against this background, we introduce the Widget Dictionary, designed for ISL avatar generation. It stores isolated, gloss-aligned signs with holistic 3D body and hand annotations, is scalable to the size of the ISLRTC dictionary [22], and continuous sentence-level animation is composed from it at inference time.

Symbolic generation systems can synthesise motion sequences directly from HamNoSys notation. For ISL, Sugandhi et al. [48] convert Hindi or English input into HamNoSys and drive a hand-crafted avatar using procedural blending. Other works generate 3D or keypoint-level motion sequences conditioned on glosses, such as the progressive-transformer family introduced by Saunders et al. [46], but to the best of our knowledge none of these target ISL. Recent sign language generation research has increasingly moved toward data-driven neural pipelines. A common formulation decomposes the task into text-to-gloss, gloss-to-pose, and pose-to-video stages, while some newer systems attempt to generate pose directly from text [20, 50, 51]. Transformer-based models such as progressive transformers improved continuity by producing variable-length sign pose sequences rather than concatenating isolated signs. GAN-based methods further improved photorealism in the final rendered videos [45, 47]. More recently, diffusion-based approaches have been applied to gloss and text-conditioned sign production, including 3D SMPL-X motion generation and attribute-aware conditioning. The goal of these approaches is improving naturalness, articulatory detail, and semantic fidelity [2, 21, 50]. Such avatar approaches are often hand-crafted and produce robotic, unnatural movement precisely because they are not captured from real human motion. Earlier notation-driven avatar systems built around a predefined protocol and character follow the same pattern [3, 11, 12, 15, 58]. To the best of our knowledge, the Widget Dictionary (Section 3.1) is the first source of ISL 3D avatar motion built from real, captured signing. Our pipeline also performs sentence-level motion synthesis directly in the browser.

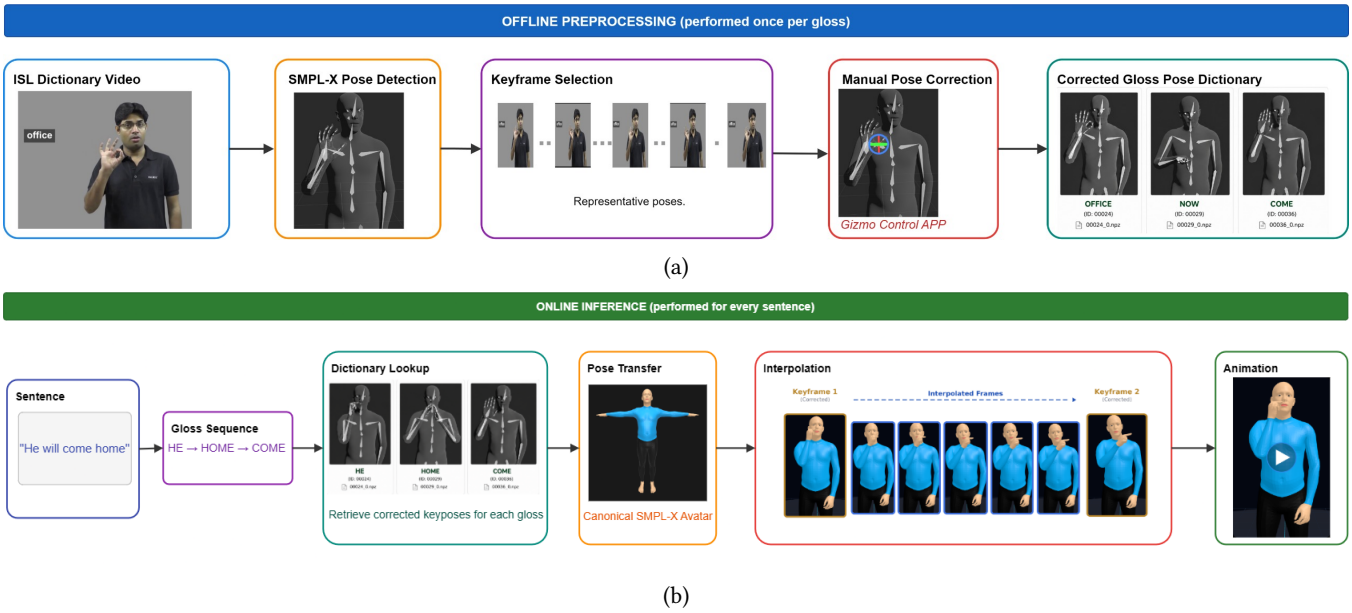


Figure 3: The SignPlug pipeline. (a) Offline dictionary construction, each ISL dictionary video is processed by SMPL-X pose detection to recover an initial 3D motion sequence, sparse keyframe selection reduces this to a small set of linguistically salient keyposes, manual pose correction fixes finger and wrist errors in these keyposes, and the corrected result is stored as a reusable entry in the Widget Dictionary. (b) Online sentence synthesis, an input sentence is converted to a gloss sequence via LLM based gloss extraction, the corrected keyposes for each gloss are retrieved from the dictionary, and the retrieved poses are retargeted onto a single canonical SMPL-X avatar and interpolated for real time playback.

3 SignPlug Pipeline

We first convert the input sentence into an ISL gloss sequence using a large language model (LLM), following recent LLM-based gloss generation [16]. Any LLM can be used for this step.

3.1 Widget Dictionary

For each gloss in our vocabulary, we source a monocular reference video of a signer performing the corresponding ISL sign from the ISLRTC Dictionary [22]. Using existing dictionary style video is a deliberate design constraint. It allows the Widget Dictionary to be extended to new words simply by recording or sourcing one additional clip per gloss, without requiring a controlled motion capture setup. Monocular capture is also a practical choice more broadly, since sign language generation has traditionally required expensive capture setups or substantial manual intervention, while recent work shows that plain RGB input can drive avatar oriented sign reconstruction pipelines [19, 31]. Dictionary style isolated recordings are likewise compatible with motion first lexicon construction, where a single recording per sign can be stored as a reusable, signer independent motion unit [19]. We call the result of this construction process the Widget Dictionary.

Each reference video is processed using SMPLer-X [6], an expressive human pose and shape estimator that regresses SMPL-X [42] body, hand, and face parameters from monocular RGB input. SMPL-X is well suited to this pipeline because it jointly models body pose, hand pose, and facial expression in a unified parametric form from monocular RGB input [29, 30, 42]. This whole body formulation matters for sign language specifically, since manual and non-manual channels must be modeled together rather than as isolated hand trajectories [29, 30]. We use the SMPLer-X ViT-H model checkpoint, `smpler_x_h32`, to estimate frame wise SMPL-X pose parameters. The recovered rotations, shape coefficients, and expression parameters are instantiated with the SMPL-X male template to obtain a mesh and skeleton for each video frame. This automatic fitting step converts each dictionary video into an initial 3D motion sequence, which is then refined through sparse manual correction.

ISLRTC Dictionary videos contain substantial temporal redundancy. Within a single sign, many consecutive frames differ only marginally, and the sign's meaning is largely determined by a small number of distinctive poses. We therefore select a sparse set of representative *keyposes* per gloss, marking the start, end, and any intermediate frames that capture a meaningful change in handshape, orientation, or location. This reduces both annotation effort and storage, while preserving the pose structure needed for avatar rendering. Sparse keyframes have also been adopted as anchors for learned sign language production [34].

Automatic SMPL-X fitting from monocular video is efficient, but it is not always linguistically reliable. In many glosses, the most frequent errors occur in finger articulation and wrist orientation, and these errors can alter the meaning of a sign. This is a known weakness of monocular whole body estimation, especially for hands and wrists under occlusion, truncation, and blur [6, 17, 56].

More broadly, monocular 3D pose estimation has advanced substantially, but depth ambiguity, occlusion, blur, and truncation remain persistent failure modes in real-world settings [17, 24]. Prior sign language work makes the same point more directly, showing that current general purpose 3D pose estimators do not perform satisfactorily for sign language generation, partly because signing motion differs substantially from the motions seen in standard human pose datasets [31].

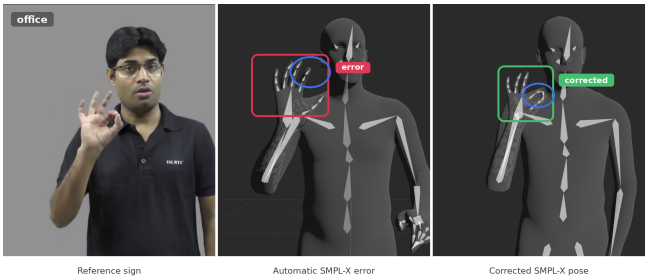


Figure 4: Human guided correction of the gloss "office". (a) The ISLRTC dictionary video provides the reference sign. (b) Automatic SMPL-X reconstruction introduces a finger curl error that changes the intended handshape. (c) The manually corrected pose, matching the reference sign, is stored in the Widget Dictionary.

Figure 4 illustrates this issue for the gloss *office*. Figure 4(a) shows the source video, which provides the intended hand configuration and finger curl. Figure 4(b) shows the automatically recovered SMPL-X pose, which articulates the fingers incorrectly. Figure 4(c) shows the corrected pose after annotation. Such errors are not cosmetic, since sign language meaning is determined by handshape, orientation, location, movement, and non-manual cues, so a small pose error can change the perceived sign or make the avatar motion unintelligible. Small errors of this kind are not cosmetic in sign language more generally, since sign meaning depends on the combined information carried by these channels [10, 29, 30]. This motivates explicit human correction rather than accepting the automatic reconstruction directly. We therefore manually correct only linguistically salient keyposes, since monocular whole body reconstruction remains vulnerable to hand, wrist, and occlusion errors that are disproportionately harmful for sign interpretation [17, 31, 56].

Rather than requiring frame by frame manual animation, we manually correct only the sparse keyposes selected above using a dedicated interactive tool (*Gizmo*), shown in Figure 5. The interface presents the source ISLRTC Dictionary video alongside the reconstructed SMPL-X avatar, allowing annotators to compare the avatar pose against the human reference. For a selected keypose, the annotator selects a bone from the SMPL-X skeleton and adjusts its rotation about the local X, Y, and Z axes using an on screen rotation gizmo, while camera focus presets for the full body, left hand, and right hand allow detailed inspection. Correction targets the manual articulators, namely body, arm, wrist, and finger joints. Facial parameters are kept as estimated by SMPLer-X and are not corrected.

To accelerate correction across a sequence, the tool provides a *Transfer Finger Poses* operation, which propagates a bone's corrected rotation to any target frame in the sequence rather than requiring the annotator to re-correct it independently. Annotators can also revert a frame when necessary and save the corrected keyframe once satisfied. This workflow reduces annotation effort while preserving the articulatory details needed for sign comprehensibility.

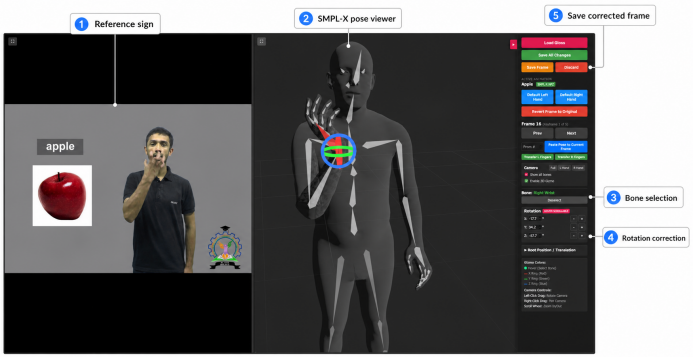


Figure 5: The Gizmo annotation interface for sparse SMPL-X correction. The reference sign video and the reconstructed SMPL-X pose viewer are shown side by side, with controls for bone selection, per-axis joint rotation, camera focus presets, frame navigation, and saving a corrected keypose.

To ensure consistency across the dictionary, all ISLRTC source videos are preprocessed to a uniform 25 FPS. Prior to pose estimation, each detected signer is cropped and resized to a fixed canonical patch. This excludes scale, camera distance, and frame position from the extracted joint rotations at the input stage, rather than removing them afterward. The pipeline estimates local, relative SMPL-X pose parameters, namely body, hand, and jaw rotations, rather than modeling raw pixel intensities. It additionally estimates per-frame shape (β) and camera translation (t) parameters, but these are discarded at the retargeting stage. Local joint rotation matrices are mapped onto a single canonical SMPL-X avatar with fixed, neutral shape coefficients and no translation offset. This IK/FK style retargeting preserves the original kinematic hierarchy and normalizes all body proportions, including hand dimensions, across signers regardless of the shape or camera translation values estimated during capture. This separation of motion from signer identity follows recent signer independent lexicon formulations for sign synthesis [19].

The output is the Widget Dictionary $\mathcal{D}$ that maps each gloss to a corrected SMPL-X motion sequence. Because each entry is stored independently of any particular sentence, dictionary entries are fully reusable. Once a gloss has been reconstructed and corrected, it can be composed into many sentences without repeating pose estimation or correction.

After correction and retargeting, each gloss is stored as an avatar ready SMPL-X motion asset with its gloss label and timing metadata. The renderer fills short intervals between corrected keyposes during playback, but this is a supporting step rather than the core contribution. The resulting assets form the Widget Dictionary used for real time sentence level avatar generation.

3.2 Runtime Sentence Level Avatar Synthesis

Each predicted gloss g_i is looked up in the Widget Dictionary to retrieve its corrected keypose sequence K_{g_i}. Because retrieval is a simple key lookup against precomputed assets, this step incurs negligible computational cost regardless of sentence length.

Because dictionary entries store pose parameters rather than rendered images, the retrieved motion can be applied directly to the canonical avatar. This keeps the signer identity fixed across all glosses in a sentence and avoids visual discontinuities caused by switching between source videos.

For each gloss, we retrieve its stored sparse keyposes, where each keypose is indexed by its original frame number in the source motion sequence, for example frame 3 or frame 10. To assemble a sentence level animation, transition frames (10 per boundary) are inserted between consecutive glosses, with a predefined temporal offset separating keyposes within a gloss. These transitions are generated using per-bone SLERP (Spherical Linear Interpolation), which produces smooth, natural looking motion between poses. This approximates the broader discrete to continuous transition problem studied in recent sign generation pipelines, which frame the task as transforming discrete gloss level units into temporally coherent continuous motion trajectories [19]. Our method performs this transition with interpolation rather than a learned transition model, so it is best described conservatively as a retrieval based composition strategy for smooth cross gloss playback rather than fully natural co-articulation learning [21]. The retrieved motion sequences are thus assembled into a single animation by preserving each gloss's internal temporal structure while adding a small number of transition frames between consecutive glosses to avoid abrupt visual discontinuities. Because the system retrieves precomputed SMPL-X gloss motions rather than synthesizing motion through a neural model, the retrieval based runtime cost remains low.

3.3 Scalability

The dictionary design allows the system to scale to larger ISL vocabularies with only linear, one time cost per new gloss. Adding a word requires sourcing or recording a dictionary style video, estimating and correcting its keyposes once, and retargeting it onto the canonical avatar, after which it is immediately available to every downstream sentence. Lexicon style storage of this kind requires only one recording per gloss, and vocabulary recombination supports generalization to novel sign combinations [19]. In principle, this makes the pipeline scalable to the full ISLRTC dictionary, since each new gloss requires only a one time monocular capture, correction, and dictionary insertion step [19, 22]. Procedural ISL systems similarly argue for scalable runtime deployment through real time, cross platform avatar pipelines, though those systems use HamNoSys and SiGML rather than retrieved SMPL-X motion assets [4]. The full pipeline is exposed through a web based translation widget in which a user selects a sentence and the corresponding avatar animation is synthesized and played back with playback and scrubbing controls, entirely client side.

4 Results & Discussions

We evaluate SignPlug along three axes that match its main design goals. First, we test whether sentence level SMPL-X avatar animation can be synthesized with low latency on the web, once an input sentence's gloss sequence is known. Second, we test whether manual pose correction makes the avatar's signing closer to the real signer's. Third, we test whether Deaf ISL signers can understand the avatar. The experiments validate practical deployment behaviour rather than compare against large neural generation models, since our method does not execute a motion generator during inference. Throughout this section we report synthesis latency separately from gloss prediction latency, since the two are handled by different, decoupled components of the pipeline (Section 3.2).

4.1 Runtime Synthesis Performance

Table 1: End to end synthesis latency and steady state playback frame rate as a function of sentence length, measured over 10 runs per length with the one time cold start run excluded. Latency is the one time cost to retrieve, retarget, and interpolate all glosses in a sentence before playback begins.

Sentence Length	Latency (s)	FPS
2 glosses	0.129	25.02
5 glosses	0.296	25.00
10 glosses	0.542	25.00
20 glosses	1.419	25.00

The Latency column reports the one time preparation cost before playback begins. This is the time to look up, retarget, and interpolate every gloss into a single ready to play animation. Latency scales gently with sentence length, from 0.129 s for two glosses to 1.419 s for twenty. Even for the longest sentence, preparation delay stays under 1.5 s. The resulting animation then plays for several seconds at native frame rate. In practice this is a brief, one time wait rather than a per frame or per gloss delay the user notices repeatedly.

The FPS column reports the steady state playback rate once synthesis completes. The ISLRTC dictionary reference videos are recorded at 25 FPS, the standard rate for this style of video capture. SignPlug sustains this rate across every tested sentence length. It reaches 25.02 FPS for a two gloss sentence and settles to 25.00 FPS from five glosses onward. The avatar animation plays at the same frame rate as the original signing footage, with no dropped frames and no visible stutter. The rendering pipeline never bottlenecks playback. The animation matches the smoothness of the original reference recordings. Beyond sustaining this native frame rate, the underlying per frame motion is free of jumps or discontinuities at gloss boundaries, as shown in Figure 7.

4.2 Effect of Manual Pose Correction

We test whether manual correction makes the avatar's signing closer to the real signer's. We embed each gloss's real signer RGB video with the Uni-Sign sign language recognition model. For each gloss, we render two avatar videos: one from the raw SMPL-X keyposes (no manual correction) and one from the keyposes corrected



Figure 6: Overview of Widget Dictionary construction, from monocular ISL dictionary videos through automatic SMPL-X estimation and sparse human correction to the Widget Dictionary supporting sentence level avatar playback.

in Gizmo, and embed both the same way. Each avatar video embedding is compared by cosine similarity with the embeddings of all real signer videos, which are ranked by this score. An avatar gloss video counts as a correct Top-k retrieval if the real signer video of the same gloss is among the k highest ranked. We evaluate over all glosses in the Widget Dictionary.

Table 2: Effect of manual pose correction on gloss retrieval over all Widget Dictionary glosses. Each avatar gloss video is matched against all real signer videos using Uni-Sign embeddings. A retrieval is correct at Top-k if the real signer video of the same gloss is among the k highest ranked. Values are in %.

Avatar videos	Top-1	Top-3	Top-5	Top-10
Raw	10.17	20.17	27.33	38.00
Pose Corrected	**35.17**	**53.50**	**62.00**	**72.00**

Table 2 shows that manual correction improves retrieval at every k. Top-1 accuracy rises from 10.17% to 35.17%, about 3.5 times higher, and Top-10 accuracy rises from 38.00% to 72.00%. The corrected avatar is therefore recognized as the intended sign far more often. This confirms that the finger and wrist errors fixed in Gizmo matter for sign identity, not only for appearance. The remaining errors are expected, since many ISL signs share similar arm movements and differ mainly in handshape or in non-manual cues that our avatar does not yet model.

4.3 Study with Deaf ISL Signers

We ran a study with 11 Deaf and hard-of-hearing ISL signers on 34 avatar sentences synthesized by SignPlug. For each sentence, participants watched the avatar video and chose the sentence it signs from four options. They then rated the sentence on a 5-point Likert scale for understandability, naturalness of motion, body posture, and hand shape. After using our live web demo, they also rated its usability.

Table 3 summarizes the results. Signers identified 88.8% of sentences exactly (332 of 374 responses), well above the 25% chance level. The avatar therefore conveys the intended sentence reliably. The Likert ratings, between 2.48 and 2.67, show that signers still

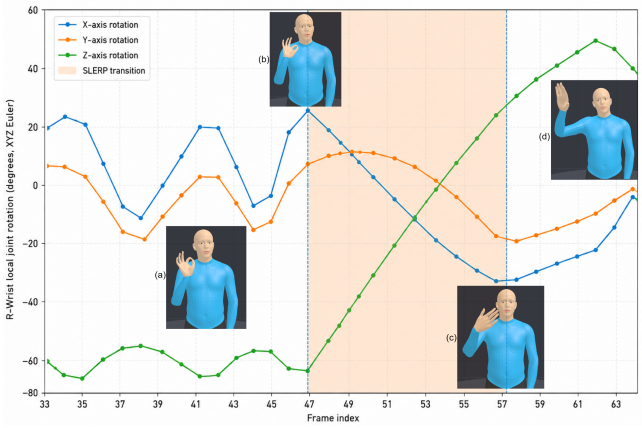


Figure 7: Smooth per frame motion at a gloss boundary. Right wrist local rotation (degrees, XYZ Euler) across the OFFICE→GO boundary (frames 33–64). The shaded band (frames 47–57) marks the SLERP interpolation span. Dashed lines mark boundary frames. (a) OFFICE mid gloss (frame 35). (b) Last keypose entering transition (frame 47). (c) First keypose exiting transition (frame 57). (d) GO mid gloss (frame 64).

Table 3: Study with 11 Deaf and hard-of-hearing ISL signers on 34 avatar sentences. Identification is the share of responses that picked the correct sentence out of four options. Ratings are means on a 1–5 Likert scale.

Measure	Result
Sentence identification (four options)	**88.8%** (332/374)
Understandability	2.67
Naturalness of motion	2.48
Body posture	2.50
Hand shape	2.61
Web demo usability	3.00

found the motion less natural than human signing. Their feedback explains why. They stressed the importance of facial expressions, and noted that some signs change form in context, for example GO in "he is going home." A dictionary of isolated signs cannot yet capture such changes. These findings motivate our future work on context-dependent sign forms and facial expressions.

5 Conclusion

Building on holistic 3D mesh reconstruction, we present SignPlug, a scalable pipeline for ISL avatar generation that runs on consumer grade compute without runtime neural inference. Our Widget Dictionary (Section 3.1), built via automatic SMPL-X estimation and sparse human correction, yields reusable motion assets at a linear cost per gloss. Runtime experiments show that sentence level animations synthesise with sub second to low second latency and play back at the native 25 FPS of the source recordings, entirely client side in the browser. A study with 11 Deaf and hard-of-hearing ISL signers shows that they identify 88.8% of avatar sentences correctly, and manual correction raises Top-1 gloss retrieval from 10.17% to 35.17%.

Limitations and future work. This work targets manual articulators, namely body pose and handshape. Non-manual markers such as facial expressions carry grammar and meaning in ISL, and our pipeline does not yet model them. Our study with Deaf signers confirms this, and also shows that some signs change form in context. Future work will add facial expressions and context-dependent sign forms, and will automate the sparse correction step through learned pose refinement. We hope this work, and the pipeline behind it, encourage the community to build stronger, more expressive, and more widely deployable tools for ISL accessibility on the web.

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Supplementary material. Project page with videos: https://sharif-commits.github.io/SignPlug/